

*Preventing young sudden cardiac deaths through
awareness, screening and research
and supporting affected families*

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Director of Screening
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Marc-Vivien Foé (1 May 1975 – 26 June 2003)



Fabrice Muamba (17 March 2012)



Sudden deaths in elite athletes

leading up to the 2012 Olympics



CRY offered testing to all potential Olympic Athletes leading up to 2012

- approx 1,500.
- launched in 2007
- concluded in 2012

Are these rare tragic events, or are they events that touch us all.

Deaths in elite athletes between 2007 and 2012

In 2007 GB Triathlon

In 2008 GB squash (TASS)

In 2009 GB Rowing

In 2011 GB Hockey U16

2013 GB Squash U18/ ASA Swimming U18

The risk is greater in sports, but the majority of deaths are not in elite athletes

Young Sudden Death

most deaths are not in elite athletes

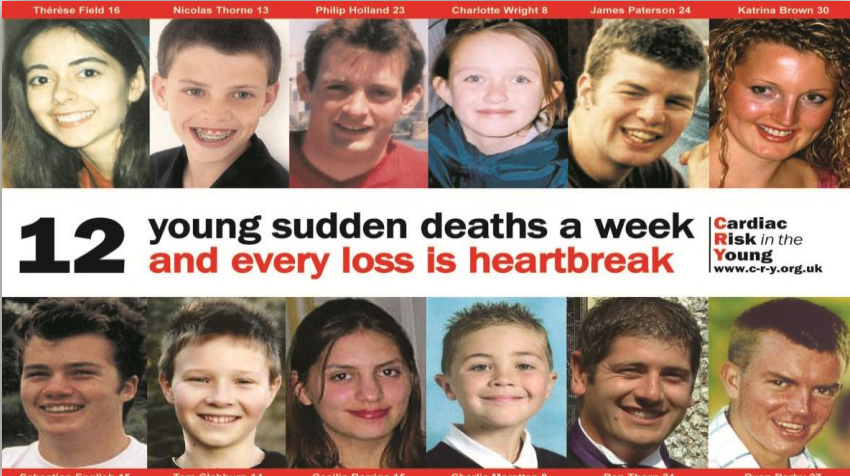
Claire Squires (London Marathon 2012)



The statistics on Young Sudden Cardiac Death

The number of apparently fit and healthy young people believed to be dying every week in the UK of undiagnosed heart conditions

1995	Only 1 a week	<i>Anecdotal evidence</i>
2000	4 to 8 a week	<i>Anecdotal evidence</i>
2004	At least 8 every week	<i>Anecdotal evidence</i>
2008	At least 12 every week	<i>published data (ONS*)</i>



12 young sudden deaths a week
and every loss is heartbreak

Cardiac Risk in the Young
www.c-r-y.org.uk

Thérèse Field 16 Nicolas Thorne 13 Philip Holland 23 Charlotte Wright 8 James Paterson 24 Katrina Brown 30

Sebastian English 16 Tom Claburn 14 Cecilia Barriga 15 Charlie Morettes 8 Ben Thorn 24 Ryan Darby 27

Most common causes of death – age 20-34 - England & Wales ONS 2009 <https://www.youtube.com/watch?v=M-onJZRZ8Zs>



Deaths in age group 20-35						
	Male (4095)		Female (1824)		Total (5919)	
Suicide	860	21%	219	12%	1079	18%
Land transport	532	13%	109	6%	641	10%
Accidental poisoning	409	10%	91	5%	500	8%
YSCD definite (exc. dissection*)	248	6.1%	87	4.8%	333	5.6%
YSCD definite (inc. dissection*)	262	6.5%	95	5.2%	357	6.0%
YSCD definite (exc. diss* + possible YSCD**)	287	7.1%	104	5.7%	394	6.7%
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YSCD data - Papadakis et al., 2009. Europace Magnitude of young sudden cardiac death in the young: a death certificate-based review in England and Wales

* dissection was excluded as a cause of death from the original paper

** possible YSCD - include 20% of epilepsy

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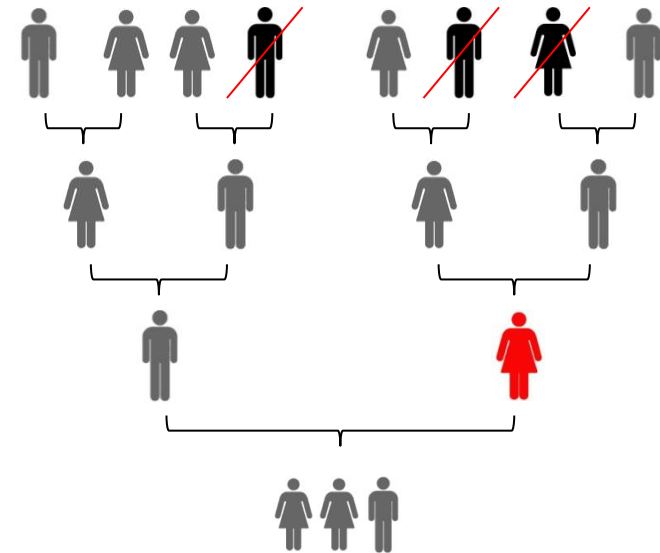
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Young Sudden Cardiac Deaths

the impact



- Since CRY started (1995) there have been over 10,000 young sudden deaths in the UK
- The deceased will lose up to 40-60 years of life.
 - 10s of 1,000s of years of life lost **EVERY YEAR**
- The impact on the immediate family (>3,000/yr)
 - siblings,
 - parents,
 - grandparents and
 - sometimes children
- Wider family (niece nephew, uncle, aunt....)
- The intimate and wider social networks
 - The close friends / partners
 - Class mates at University/school, workmates, team mates
- Wider community (Fabrice Muamba / Claire Squires)



Medical Terminology

an overview of the conditions that cause YSCD



- Structural conditions
 - Cardiomyopathies (HCM, ARVC, DCM)
- Electrical conditions
 - Ion Channelopathies
 - Brugada
 - Long QT
- inherited /genetic conditions (50% of cases)
 - Problems with Genetic testing
 - The more we know – the more we don't know
- Congenital conditions (WPW/Coronary Artery Anomalies)
- Acquired conditions
 - Myocarditis
- Commotio Cordis
- Premature Coronary Artery Disease
- **YSCD vs. SADS**

CRY aims to reduce the incidence and impact of young sudden cardiac death by:

Saving young people before a tragedy occurs
Supporting people after a tragedy occurs

Awareness



Support



Screening



Research



Raising Awareness *with* the General Public, Medical and Sporting Communities



Raising Awareness saves young lives

1. Symptoms that increase risk
2. Family history of young sudden death
3. Those at risk who have no symptoms
www.testmyheart.org.uk



Raising Awareness

Saves young lives when there are symptoms

Symptoms that put you at risk

chest pain (exercise related) 
passing out (syncope) 
breathlessness
palpitations
dizziness



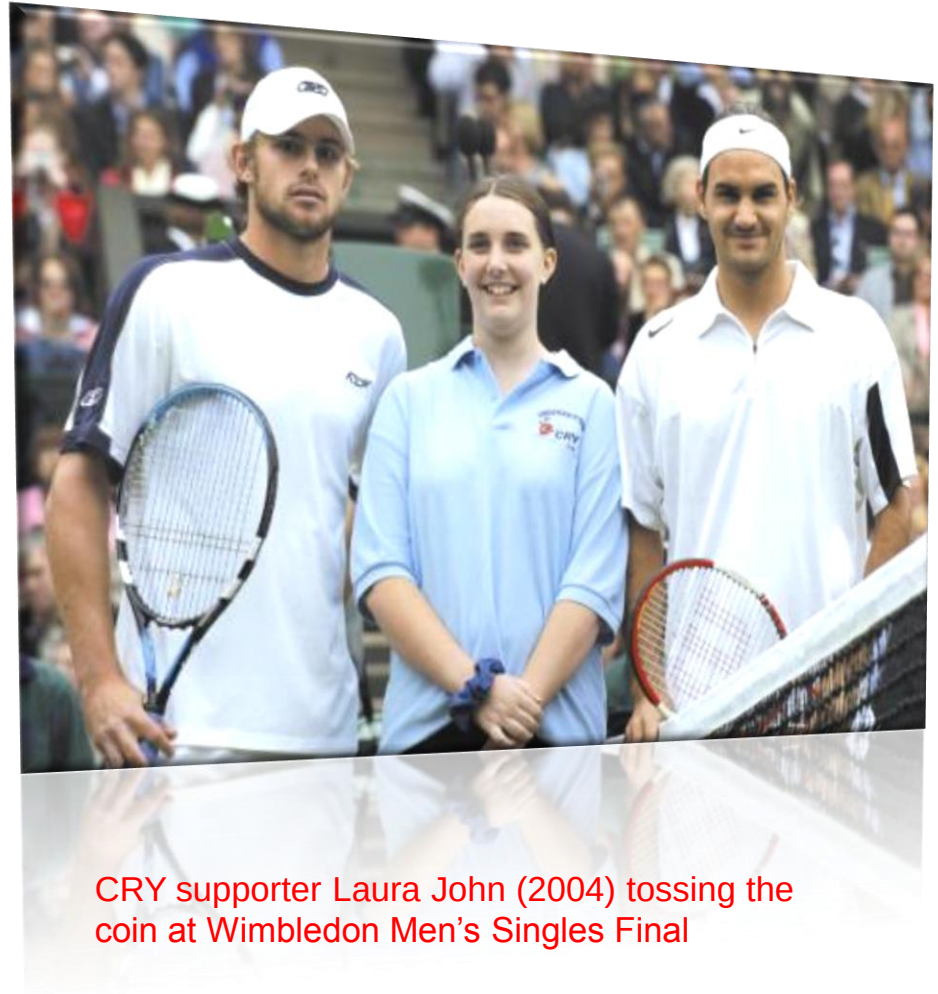
Raising Awareness

Saves young lives when there is a family history

Family history of young sudden death

After a young sudden death the family members need to know;

- where to go for emotional support
 - Talk to CRY Founder and CEO
 - Support
 - Bereavement
 - Diagnosed / *myheart*
- the cause of death
 - Unascertained/natural causes
 - Expert pathology
- if others are at risk
 - Genetic conditions
 - Expert cardiology referral



CRY supporter Laura John (2004) tossing the coin at Wimbledon Men's Singles Final

Support through expert pathology

CRY Centre for Cardiac Pathology (CRY CCP) *Established in 2008*



The importance of correct pathology cannot be overstated as it gives families the opportunity to obtain valid answers to the cause of death and to quantify the risk posed to other family members.



Unexplained sudden death often recorded as 'natural causes'

Fast-track expert cardiac pathology service
2 weeks instead of up to 24 months

Service lead by world expert Dr Mary Sheppard

Pathology informs clinical tests (e.g. HCM vs. Brugada)





Professor Sanjay Sharma heads up a team of specialist doctors based at St. George's Hospital, London.

Services for:

- affected families,
- competitive athletes *and*
- the general population

CRY funds the machinery, 7 doctors / research fellows, a physiologist, and a nurse

At these fast-track clinics families will be seen

- Within a few weeks of referral, after the sudden death of a young family member
- Have all tests on the same day
- Be seen together as a family unit

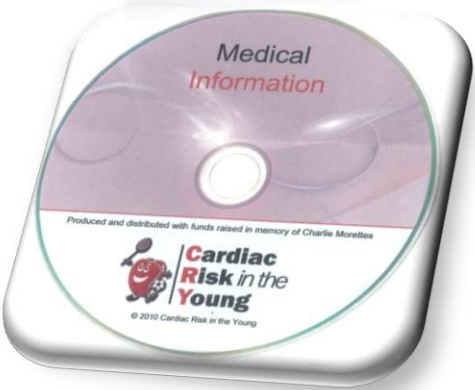
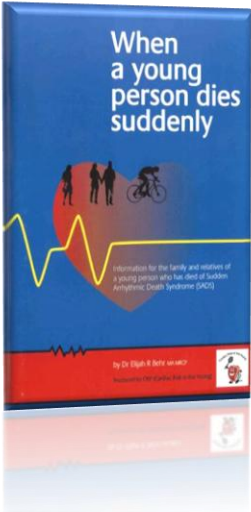
The CRY Centre for Sports Cardiology (CRY CSC) provides a specialist service for expert clinical cardiac diagnoses of elite athletes.

London Screening clinic at weekends for the general public

Support for families *through* Medical Information



CRY's YouTube channel:
www.youtube.com/user/cryvideos



Following a young sudden cardiac, death families often seek emotional support and further information.

Individual Support - CRY's Bereavement Support Network



20 volunteers with experience of a similar tragedy undertaken CRY's accredited counselling training to help others adjust.

Bereavement Supporters complete a two year BAC accredited Counselling Skills and Theory course.

Group Support - CRY's Bereavement Support Days

National bereavement support days in **Birmingham** specifically for siblings, partners, mothers and fathers.



Support for Those Affected *through* CRY's *myheart* Network



myheart
Supporting young people with cardiac conditions

What is *myheart*?

for young people (age 15 to 35) diagnosed with a cardiac condition

Meetings

meet with other young people with similar conditions

talk about your experiences in a supportive environment

ask a specialist questions in an informal setting

understand what but also why some things need to be avoided

Help from *myheart* members

contact others with similar conditions

help adjust to limitations (exercise/lifestyle)

Newsletter and website

regular newsletters

www.myheart.org.uk website

All of the support services CRY offers are free



Raising awareness *of the importance of cardiac screening for all young people*

The simple way to diagnose most cardiac abnormalities is a quick, painless and affordable procedure called an electrocardiogram (ECG).

80% of young sudden cardiac deaths are asymptomatic

International recommendations that screening should be in place prior to participation in sport, in Italy this has been law since 1982

The majority of young sudden deaths do not occur in elite athletes, they are at grass roots

89% reduction in young sudden deaths in Italy when screening implemented

www.testmyheart.org.uk
[#testmyheart](https://twitter.com/testmyheart)



CRY's screening service is available to any person aged between 14 and 35.

Cardiac screening

all young people should have the opportunity to be tested

1 in 300 people have potentially life threatening conditions

1 in 100 have less serious cardiac abnormalities

A wise man once said “we each know a 1,000 people”,

if we tested 1,000 14 year olds

3 - identified with potentially life threatening conditions
(WPW, HCM, Long qt)

10 - less serious conditions
problems could be prevented if identified, monitored and corrected

Most large organisations (+1,000) will have had a member of staff directly affected by YSCD

All large organisations will have staff with a potential risk of YSCD

UNUM – 900 staff - Father lost his son, brother lost his sister and cousin lost his nephew.



All young people should have the choice to be screened

Why not rely on AEDs?



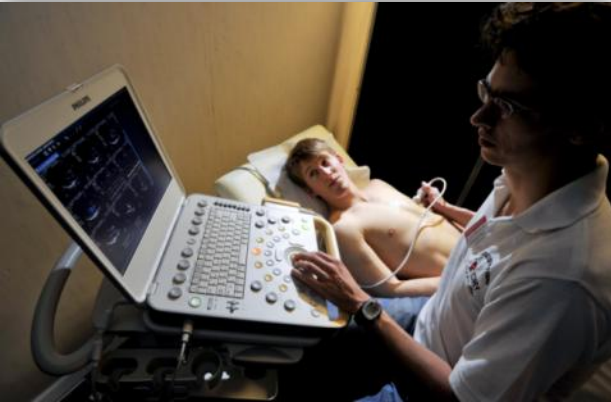
AEDs

- CRY's position is to prevent the cardiac arrest in first place
- AEDs important – role of BHF/other charities
- Majority who benefit will be over age of 50
- AEDs work with screening to prevent YSCDs (e.g. CAA / Comotio Cordis / Acquired)
- 80% of SADS occur at home or in sleep

Mandatory Screening

- Every young person should have the choice to be tested, not to be mandated for the general population
- Testing should be offered as soon as possible after puberty (age 14+)
- If money is no object ultrasound also recommended with the ECG.
- The choice of Governing bodies to mandate for their athletes (paid/sponsored players)

Cardiac Screening for Elite Athletes in the UK



Sport itself does not lead to cardiac arrest but can trigger a young sudden death by aggravating an undetected abnormality.

The European Society of Cardiology (ESC) and International Olympic Committee (IOC) recommend cardiac screening for any young person participating in competitive sport.

In countries such as Italy, screening for those participating in representative sport is mandatory. This is also the case in some professional sports (UEFA, FIFA, UCI)

For some professions (e.g. pilots) cardiac testing is also mandatory.

Philips provided CRY: 2 CX50 Portable ECHO machines, 1 ie33 3D echo machine & 4 ECG machines; plus funding to test all elite GB athletes leading up to the 2012 Olympics.

CRY tests many of the leading UK athletes, including;

GB Rowing, GB Cycling,
GB Swimming, GB Diving,
GB Boxing, UK Athletics,

Premiership Football Teams
(e.g. Manchester City,
Tottenham),

England Cricket,
The Lawn Tennis Association,
SKY Cycling.

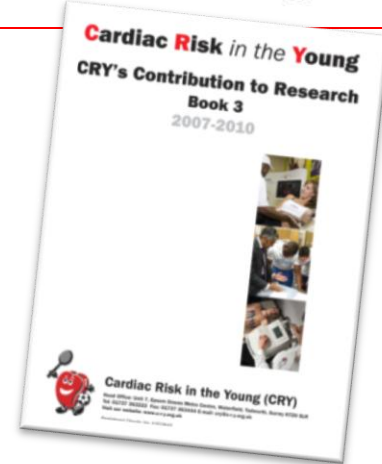
CRY tests all Aviva Premiership
Rugby players, England
representative and Regional
Academy players above the
age of 16

CRY's screening programme not only identifies people at risk, but also provides data for vital research into the prevalence and management of cardiac abnormalities.

The findings are published in reputable, peer-reviewed journals and our guidelines are internationally recognised.

CRY was the first organisation to:

- identify the upper limits of wall thickness and cavity size in British athletes;
 - characterise cardiac dimensions in adolescent athletes – knowing how to differentiate pathology (disease) from physiology (normal because of exercise) is vital for diagnosis;
 - characterise ECG changes in athletes in a document that is now the blueprint for the European Society of Sport Cardiology;
 - look at cardiac adaptation for Caribbean athletes, as they differ from Caucasian athletes in the way they adapt to exercise.
- CRY has also been pivotal in identifying the prevalence of conditions such as hypertrophic cardiomyopathy (HCM) in athletes.



- **Summer Science Exhibition at the Royal Society of Medicine**
- **ESCel videos**
- **New CRYteria – ESC screening guidelines - update on Seattle criteria**

Research

Screenings

Support NHS services



Broaden expertise in young sudden cardiac death to other UK hospitals

CRY have funded 14 research fellows, currently funding a further 7 fellows.

Creating an infrastructure of specialists throughout the UK

CRY's 9th International Conference at The Cavendish Conference Centre, London (2014)



www.c-r-y.org.uk/conferences.htm

Friday 9th October 2015

Sports Cardiology: From Theory to Practice

Organised in collaboration with the Section of Sports Cardiology of the European Society of Cardiology

Speakers include:

Professor Mats Borjesson (Sweden),
Professor Domenico Corrado (Italy),
Professor Hein Heidbuchel (Belgium), and
Professor Sanjay Sharma (UK).

This annual event attracts speakers and delegates from around the globe, including many of the world's leading experts on sports cardiology and young sudden cardiac death.

There is no charge to attend either or both conference days.

Uncertainty

and the role of specialist units and expertise



Uncertainty with inherited conditions

- *genetic risk vs. clinical risk*
- *tests do not always predict outcomes*
- *athletes heart vs. diseased heart*
- *electrical problems - cause unknown unless identified in 1st degree relative*
-

If the cardiologist treating me is uncertain I would want to see a cardiologist with more knowledge, more experience

Uncertainty of an expert is very different from uncertainty due to lack of knowledge

Young Sudden Cardiac Death is “rare” - death in young people is “rare”

The conditions that cause Young Sudden Cardiac Death are “common”

Cardiac Screening

the importance of expertise and follow up


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Tottenham Hotspur liable for player brain damage



Radwan Hamed (left) signed for Spurs' youth team without knowing he had had an abnormal ECG

A High Court judge has ruled Tottenham Hotspur breached its duties to a 17-year-old player who suffered cardiac arrest in his first game for the club and was left brain damaged.

Radwan Hamed collapsed during the youth team game in Belgium in August 2006.

The way risk of YSCD is currently presented



confusion
incorrect policy
false reassurance

*To say that 800 athletes need to be stopped
from playing competitive sport to prevent
1 sudden death is wrong*

Diagnosis, risk stratification and management
require an accurate evaluation
of the incidence of YSCD



To think these events are
as rare as being killed by lightning
is not only wrong but dangerous

Physical activity in adolescent and adults with congenital heart defects; individualised exercise prescription

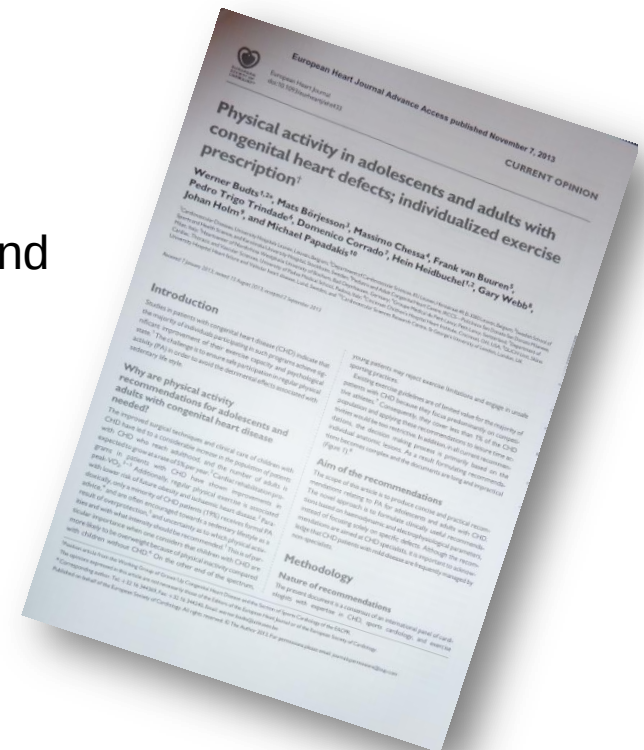
Budts, Borjesson, Chess, Buuren, Trindade, Corrado, Heidbuchel, Webb, Holm and Papadakis EHJ 2013

Consensus guidelines

Reassurance to patients and physicians

Individualised exercise prescription to
maximise the benefit for health and
minimize the cardiovascular risk

Improve research base



Overprotective!

Good for safety - not great for street credibility!



Moderate physical activity is important

Many cats have hypertrophic cardiomyopathy (HCM)

Moderate exercise would be beneficial!



© London Media

There needs to be a balance

“I want to run the London Marathon”

Some physical activity is not normal

London Marathon

Iron man triathlon

Marathon de Sables

NO NO NO!



Cardiac conditions are one of the most common causes of death in young people

The impact of a YSCD has a devastating impact on thousands of people every year

Cardiac conditions that are potentially life threatening are common

Cardiac conditions in young people can be identified through screening

Risk of sudden death is significantly reduced once a young person is identified

With the correct support young people with conditions can live “normal” lives

Thank you for listening



/CardiacRiskintheYoung



@CRY_UK



/cryvideos

